From the Equivalence Theorem to Photon-Path Cumulant Fitting

Derivation, assumptions, literature context, and application to OCO-2 spectra

OCO-2 cloud-proximity analysis: technical report

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Abstract

This report derives the spectral transmittance fit used in the OCO-2 cloud-proximity analysis from the radiative-transfer equivalence theorem. In a narrow band such as the oxygen A band, the continuum scattering properties are assumed to be effectively wavelength independent. A single conservative scattering calculation then defines an ensemble of photon trajectories, and gas absorption reweights each trajectory by its Beer–Lambert survival probability. Under the additional scalar-path approximation, the normalized radiance is the Laplace transform of a wavelength-independent photon path-length distribution. Expanding its logarithm gives the fitted mean path enhancement, path variance, and higher scaled cumulants. We state the assumption required at every step, distinguish the underlying path ensemble from the absorption-selected paths actually detected in strong and weak channels, and explain how the classical literature maps onto the equations and the current implementation.

1 Purpose and scope

The practical question is whether a resolved absorption spectrum can be used to infer information about the photon paths created by clouds, aerosols, the surface, and three-dimensional transport. The foundational result is the equivalence theorem of Irvine (1964), developed for high-resolution atmospheric calculations by Partain et al. (2000) and applied to molecular-line absorption and the O_2 A band by Stephens and Heidinger (2000) and Heidinger and Stephens (2000). High-resolution measurements have subsequently been used to infer photon-path distributions or their moments (Funk and Pfeilsticker, 2003; Min et al., 2004; Scholl et al., 2006).

The present project does not invert a complete path-length probability density function (PDF). It fits a truncated expansion of log transmittance against modeled slant optical depth, separately for O_2 A, weak CO_2, and strong CO_2. The goal of this report is to show exactly when the fitted coefficients have a photon-path interpretation and when they should instead be called effective spectral-shape coefficients. This report is the expanded theoretical companion to `FITTING_DERIVATION.md`; both documents follow the same project sequence: conservative trajectory ensemble, Beer–Lambert weighting, scalar-path reduction, Laplace transform, continuum normalization, log-cumulant expansion, and exact linear fitting.

2 Objects that must be kept distinct

Let Γ denote a complete photon trajectory from solar illumination to the detector, including every scattering and surface interaction. Three related distributions appear in the argument:

1. $P_0(\Gamma)$ is the trajectory distribution in the corresponding conservative-scattering problem, before the target gas absorption is applied.

2. $p_0(\ell')$ is the distribution of a scalar normalized path coordinate ℓ' induced by $P_0(\Gamma)$.

3. $p_{\text{det}}(\ell' \mid \tau)$ is the distribution among photons that survive absorption and reach the detector at optical depth τ.

The equivalence theorem requires the first distribution to remain unchanged when absorption is introduced. It does *not* require the detected-path distribution to be the same in weak and strong absorption channels. Strong absorption preferentially removes long paths, so a change in $p_{\text{det}}(\ell' \mid \tau)$ is an expected prediction of the theorem, not evidence against it.

3 Equivalence theorem at the trajectory level

Consider a narrow spectral interval and a fixed illumination–view geometry. Let $I_{0,\lambda}$ denote the radiance for an otherwise equivalent reference problem in which the target molecular absorption is absent. The absorption optical depth accumulated along trajectory Γ is

$$A_\lambda(\Gamma) = \int_\Gamma \alpha_\lambda(\boldsymbol{r})\,\mathrm{d}s, \tag{1}$$

where α_λ is the local molecular absorption coefficient. The Beer–Lambert survival probability of that trajectory is $\exp[-A_\lambda(\Gamma)]$. Averaging over the conservative trajectory ensemble gives

$$\frac{I_\lambda}{I_{0,\lambda}} = \int P_0(\Gamma)\exp[-A_\lambda(\Gamma)]\,\mathrm{d}\Gamma = \mathbb{E}_0\{\exp[-A_\lambda(\Gamma)]\}\,. \tag{2}$$

Equation (2) is the general equivalence-theorem statement relevant here. It is independent of whether the scattering medium is plane parallel or three dimensional.

Stephens and Heidinger (2000) formulated the O_2 A-band problem in terms of the distribution of continuum photon optical paths. Their equivalence-theorem construction separates the continuum multiple-scattering calculation from line absorption: a Monte Carlo calculation at a nonabsorbing wavelength first provides the path distribution and continuum radiance, after which the transmission function associated with gaseous absorption is integrated over that distribution. The same stored path statistics can therefore generate a complete high-resolution line spectrum as the line-to-continuum absorption ratio is varied. They demonstrated this numerically by comparing spectra reconstructed from the path distribution with direct multiple-scattering calculations, finding essentially identical results for their test case. They also applied the construction grid point by grid point to horizontally heterogeneous cloud fields: three-dimensional transport changes the path distribution, but does not alter the equivalence-theorem weighting once that distribution is known. This is the precise sense in which equation (2) is geometry independent. Their subsequent O_2 A-band application showed that the line-shape response carries information on cloud optical depth, phase function, cloud pressure and pressure thickness, and surface albedo (Heidinger and Stephens, 2000). For this project, the key inheritance is narrower: cloud geometry and 3-D transport are encoded in $P_0(\Gamma)$, while wavelength-dependent O_2 absorption interrogates that ensemble.

Partain et al. (2000) developed generalized, computationally useful forms of the same theorem for line-by-line atmospheric radiative transfer. Rather than repeating a multiple-scattering solution at every spectral point, their method accumulates probability distributions of photon histories during a reference scattering calculation and applies spectral gas, particle, and surface absorption afterward. Use of geometrical path information permits different vertical gas profiles and multiple homogeneous layers to be treated, while joint path/scattering statistics extend the bookkeeping to nonconservative particles and surface absorption. The complete joint distribution is memory-intensive, so they also introduced a reduced PDF approximation; later analysis noted that compressing the full trajectory information can introduce absorption bias. This distinction is directly relevant here. The present project does not retain the complete layer-resolved or joint PDF used by the generalized theorem. It makes the stronger reduction $A_\lambda(\Gamma) \simeq \tau_\lambda \ell'(\Gamma)$ and fits a one-dimensional marginal distribution. Partain et al. therefore support the trajectory-level starting point, while also showing what information is discarded by the project's scalar-path approximation.

Assumptions at this step.

1. Introducing target-gas absorption does not change the geometry or the continuum scattering phase function, extinction, or boundary conditions.
2. Inelastic processes that change photon wavelength, such as Raman scattering, are negligible or separately corrected.
3. The reference radiance $I_{0,\lambda}$ represents the same scattering problem without the target absorption. A continuum estimate that contains unmodeled surface or instrumental spectral structure violates this identification.

4 Narrow-band reduction to one path-length PDF

For a narrow band such as O_2 A, cloud and aerosol scattering properties are often assumed to vary slowly compared with the many-orders-of-magnitude change in molecular line absorption. This is the central spectral-separation assumption used in O_2 A-band PPDF work (Stephens and Heidinger, 2000; Min et al., 2004; Scholl et al., 2006).

The three studies play complementary roles in the present derivation. Stephens and Heidinger (2000) provide the theoretical separation: if continuum particle properties and geometry are unchanged across the line, changes in spectral radiance can be generated by applying the molecular transmission function to one continuum path distribution. Their path coordinate is naturally an optical path of the continuum scattering problem. Reorganizing a simulated A-band spectrum by O_2 optical depth makes the line-growth behavior and its dependence on the path distribution explicit. In the project equations, this motivates holding $P_0(\Gamma)$ fixed while scanning τ_λ; it does not by itself justify replacing the full trajectory-dependent absorber amount by the scalar product $\tau_\lambda \ell'$.

Min et al. (2004) connect that theory to high-resolution observations. Their instrument resolved the O_2 A band and a nearby water-vapor band with better than approximately 0.5 cm^{-1} resolution and out-of-band rejection better than 10^{-5}, properties needed because line-center leakage and the instrument slit function strongly limit inverse-Laplace information. They wrote the multiple-path Beer–Lambert relation as an integral over the absorber amount encountered along photon trajectories and treated the resolved absorption line as an inverse-Laplace problem. O_2 is advantageous because its abundance profile is well known and its A-band lines span a large range of opacity. Their comparison with the nearby water-vapor band is especially important for interpretation: the inferred mean and variance differed between the two absorbers because water vapor is vertically inhomogeneous and coupled to cloud location. Thus a common geometrical scattering ensemble need not produce the same *effective absorber-path distribution* for different gases. For this project, that result supports separate O_2, weak-CO_2, and strong-CO_2 fits and motivates calling their coefficients band-effective absorber-path cumulants rather than universal geometrical-path moments.

Scholl et al. (2006) make the vertical structure still more explicit. Their forward model used pressure- and temperature-dependent O_2 line absorption and partitioned the atmospheric path into regions above the cloud, within the cloud, and below the cloud, including an approximate cloud–surface multiple-reflection contribution. The random in-cloud path was represented by a parameterized PDF and integrated against the wavelength-dependent absorption of that region; the resulting high-resolution spectrum was convolved with the measured instrumental slit function. Although their spectral resolution and out-of-band rejection supported roughly four independent pieces of information, their primary analysis retained the first two path moments and usually adopted a gamma-type PDF. For sufficiently simple, well-layered clouds and high signal-to-noise spectra, their sensitivity analysis indicated that these moments could be recovered to approximately five percent. Their measurements were transmitted skylight rather than OCO-2 reflected radiance, so their path geometry is not transferred directly to this project. Their method nevertheless identifies the exact approximation made here: the project collapses their separate above-, in-, and below-cloud absorber paths into one normalized multiplier ℓ'. Agreement of the fitted OCO-2 coefficients with layer-resolved Monte Carlo moments would therefore be a closure test of the scalar reduction, not of the equivalence theorem itself.

To reduce equation (2) to a one-dimensional Laplace transform, make the stronger scalar-path approximation

$$A_\lambda(\Gamma) \simeq \tau_{\mathrm{geo}}(\lambda)\,\ell'(\Gamma). \tag{3}$$

Here $\tau_{\mathrm{geo}}(\lambda)$ is the molecular optical depth along the direct geometrical slant path, and ℓ' is the photon absorption-path enhancement relative to that direct path. Defining

$$p_0(\ell') = \int P_0(\Gamma)\delta\big[\ell' - \ell'(\Gamma)\big]\ \mathrm{d}\Gamma, \qquad \int_0^\infty p_0(\ell')\,\mathrm{d}\ell' = 1, \tag{4}$$

and substituting equations (3) and (4) into equation (2) yields

$$\mathcal{T}(\tau_{\mathrm{geo}}) = \int_0^\infty p_0(\ell')\exp(-\tau_{\mathrm{geo}}\ell')\,\mathrm{d}\ell' = \mathcal{L}\{p_0\}(\tau_{\mathrm{geo}}). \tag{5}$$

Thus the resolved spectrum scans the Laplace transform of a single path PDF: wavelength changes τ_{geo}, while p_0 remains fixed.

Assumptions added by the scalar reduction.

1. Every trajectory's layer-by-layer path can be represented by one multiplier ℓ'. Equivalently, scattering approximately stretches or shortens the direct slant path in the same proportion in all absorbing layers.

2. The modeled $\tau_{\mathrm{geo}}(\lambda)$ contains the relevant line strength, pressure, temperature, and absorber-profile dependence accurately.

3. The same $p_0(\ell')$ applies to all retained channels in the band. Slowly varying scattering is necessary but not sufficient: surface BRDF, polarization, aerosol properties, and instrumental response must also not introduce important channel-dependent path ensembles.

The first assumption is the most consequential. Molecular oxygen is well mixed, but its line absorption is pressure and temperature dependent. A photon spending extra distance above a cloud and one spending it near the surface can have the same geometrical length and different $A_\lambda(\Gamma)$. The general theorem, equation (2), still holds; only its reduction to the scalar Laplace transform may fail.

5 Why weak and strong channels detect different paths

Even when a single underlying p_0 is exact, absorption changes the paths represented among detected photons. Bayes weighting of equation (5) gives

$$p_{\mathrm{det}}(\ell' \mid \tau) = \frac{p_0(\ell')e^{-\tau\ell'}}{\mathcal{T}(\tau)}. \tag{6}$$

Large τ suppresses the long-path tail. Consequently,

$$-\frac{\mathrm{d}\ln\mathcal{T}}{\mathrm{d}\tau} = \mathbb{E}_{\mathrm{det},\tau}[\ell'], \tag{7}$$

$$\frac{\mathrm{d}^2\ln\mathcal{T}}{\mathrm{d}\tau^2} = \mathrm{Var}_{\mathrm{det},\tau}(\ell'). \tag{8}$$

The local slope at a strong line is therefore the mean of an absorption-selected, short-path population. This does not imply an underlying wavelength-dependent p_0. A genuine violation occurs when $P_0(\Gamma \mid \lambda)$ changes across the band, or when no common scalar $\ell'(\Gamma)$ satisfies equation (3) for all channels.

6 Surface reflection and the measured transmittance

For a Lambertian surface with reflectance γ and a direct solar normalization $F_0\mu_0$, the simplified reflected radiance is

$$\pi I_\lambda = F_{0,\lambda}\mu_0\,\gamma\,\mathcal{T}[\tau_{\mathrm{geo}}(\lambda)]. \tag{9}$$

Define the measured normalized quantity

$$T_\lambda \equiv \frac{\pi I_\lambda}{F_{0,\lambda}\mu_0} = \gamma\int_0^\infty p_0(\ell')e^{-\tau_{\mathrm{geo}}\ell'}\,\mathrm{d}\ell'. \tag{10}$$

Then $T(0) = \gamma$. In the project implementation, `compute_transmittance` evaluates the corresponding radiance/TOA-solar normalization times π, and the fitted intercept absorbs the effective zero-absorption reflectance.

Assumptions at this step. The Lambertian, single-reflection expression is an interpretation, not a full description of the OCO-2 radiance. Spectral BRDF, polarization, multiple surface–atmosphere interactions, fluorescence, unresolved continuum structure, or errors in the solar normalization can enter the intercept and the fitted curvature. Accordingly, exp(intercept) should generally be called an *effective continuum reflectance*, not an exact surface albedo.

7 From the Laplace transform to the cumulant fit

Let C_n be the ordinary cumulants of the underlying normalized path distribution $p_0(\ell')$. The cumulant-generating function is

$$K(t) = \ln\mathbb{E}[e^{t\ell'}] = \sum_{n=1}^{\infty}\frac{C_n}{n!}t^n. \tag{11}$$

Because the Laplace transform is the moment-generating function evaluated at $t = -\tau$, equations (10) and (11) give

$$\ln T(\tau) = \ln \gamma + \sum_{n=1}^{\infty} \frac{(-1)^n C_n}{n!} \tau^n. \tag{12}$$

The code uses the parameterization

$$\ln T(\tau) = b + \sum_{n=1}^{N} \frac{(-1)^n k_n}{n} \tau^n, \tag{13}$$

so the exact mapping is

$$b = \ln \gamma, \qquad k_n = \frac{C_n}{(n-1)!}. \tag{14}$$

In particular,

$$k_1 = C_1 = \mathbb{E}_0[\ell'], \qquad k_2 = C_2 = \mathrm{Var}_0(\ell'). \tag{15}$$

The first two fitted coefficients are therefore the ordinary mean and variance under the assumptions above. For $n \geq 3$, k_n is a scaled cumulant rather than the ordinary cumulant itself. This factorial distinction matters when comparing the project coefficients with standard probability notation.

The design matrix used by the project is consequently

$$\boldsymbol{X}(\tau) = \left[1, -\tau, \frac{\tau^2}{2}, -\frac{\tau^3}{3}, ldots, \frac{(-1)^N \tau^N}{N}\right]. \tag{16}$$

The model is linear in $(b, k_1, \ldots, k_N)$ and is solved by SVD least squares, with a bounded linear least-squares fallback if k_1 or k_2 would be negative. Production uses the raw, non-Savitzky–Golay coefficient set; the parallel smoothed fit is retained as a robustness product. The band orders are $N = (7, 3, 7)$ for O_2 A, weak CO_2, and strong CO_2, respectively. The lower weak-CO_2 order reflects its limited optical-depth range and the poor identifiability of high powers of τ, not a different theoretical model.

Assumptions at the fitting step.

1. The Taylor expansion about $\tau = 0$ adequately represents the retained finite-τ channels. Strong line centers may lie outside the useful convergence region even when the equivalence theorem itself is valid.

2. Channel errors and forward-model errors do not create structured residuals that the polynomial mistakes for path cumulants.

3. k_1 and k_2 are identifiable from the available optical-depth range. Positivity is physically necessary but does not guarantee that the complete fitted polynomial corresponds to any nonnegative PDF.

8 Gamma distribution as an interpretive special case

If $p_0(\ell')$ is gamma distributed with mean m and shape κ,

$$p_0(\ell') = \frac{(\kappa/m)^\kappa}{\Gamma(\kappa)} \ell'^{\kappa-1} \exp\left(-\frac{\kappa}{m}\ell'\right), \tag{17}$$

then

$$T(\tau) = \gamma\left(1 + \frac{m}{\kappa}\tau\right)^{-\kappa}. \tag{18}$$

The ordinary gamma cumulants are $C_n = (n-1)!m^n/\kappa^{n-1}$, and therefore the project coefficients obey

$$k_n = \frac{m^n}{\kappa^{n-1}}, \qquad k_1 = m, \qquad k_2 = \frac{m^2}{\kappa}, \qquad \kappa = \frac{k_1^2}{k_2}. \tag{19}$$

Under a strict gamma PDF, all higher coefficients are fixed by k_1 and k_2:

$$k_n = k_1\left(\frac{k_2}{k_1}\right)^{n-1}. \tag{20}$$

The current production fit does *not* impose equations (18) and (20). It fits the coefficients in equation (13) independently, constraining only $k_1, k_2 \geq 0$. The gamma closed form exists in `cumulant_fit.py` but is not used by the active estimator. Thus k_1^2/k_2 is a gamma-equivalent shape diagnostic, not evidence that the retrieved path PDF is gamma. Flexible PDF retrievals such as the finite-element approach of Bennartz et al. (2006) illustrate why a two- parameter gamma form can be too restrictive for multilayer or bimodal path populations.

9 When one wavelength-independent scalar PDF is inadequate

The fully stratified form of the equivalence theorem remains valid if each trajectory is described by its layer path vector $\boldsymbol{L} = (L_1, \ldots, L_Z)$. Let $p_0(\boldsymbol{L})$ be its joint distribution. Then

$$T_\lambda = \gamma_\lambda \int p_0(\boldsymbol{L}) \exp\left[-\sum_{z=1}^{Z} \alpha_{\lambda z} L_z\right] \mathrm{d}\boldsymbol{L}. \tag{21}$$

This is a multidimensional Laplace transform sampled along the wavelength- dependent vector $\boldsymbol{\alpha}_\lambda$. It reduces to equation (5) only when the layer path vectors are effectively collinear with a known reference profile, $L_z \simeq \ell' \bar{L}_z$.

This distinction connects directly to prior work. Scholl et al. (2006) used pressure- and temperature-dependent O_2 absorption and separated path contributions above, within, and below cloud layers. Min et al. (2004) compared O_2 and water-vapor path statistics and showed that absorber vertical inhomogeneity changes the inferred effective paths. Bennartz et al. (2006) demonstrated the difficulty of a single classical Laplace inversion for multilayer clouds and tested a more flexible PDF representation against Monte Carlo path distributions.

For OCO-2, a failure of the scalar approximation could appear as systematic differences between channels with similar total τ but different vertical absorption kernels. This would not disprove the equivalence theorem. It would show that equation (3), rather than equation (2), is too aggressive.

10 Connection to the OCO-2 cloud-proximity project

The project's strongest defensible statement is therefore:

> Within each OCO-2 band, the fitted coefficients are effective cumulants of an absorption-path enhancement distribution under the equivalence-theorem, wavelength-invariant-scattering, and scalar-path approximations. The first two coefficients reduce to the mean and variance of a common underlying path PDF when those approximations hold.

It is stronger than calling the coefficients arbitrary polynomial features, but more accurate than claiming an exact reconstruction of the geometrical photon-path PDF.

Table 1: Relationship between the derivation, literature, and implementation.

Concept	Literature role	Project realization and interpretation
Equivalence theorem	Irvine (1964), Partain et al. (2000), and Stephens and Heidinger (2000) separate scattering-path statistics from gaseous absorption.	The project treats channels in a narrow band as samples of one scattering-generated path ensemble evaluated at different modeled optical depths.
Narrow O_2 A band	Heidinger and Stephens (2000), Funk and Pfeilsticker (2003), and Min et al. (2004) exploit nearly constant continuum scattering and strongly varying O_2 absorption.	O_2 A is fitted independently from the CO_2 bands. The O_2 coefficients are insensitive to a real CO_2 enhancement to first order and therefore provide a useful cloud/scattering fingerprint.
Path moments	Funk and Pfeilsticker (2003) and Scholl et al. (2006) infer PPDFs or their first two moments from high-resolution spectra.	k_1 and k_2 are used as mean-path and path-variance features, but should be described as band-effective quantities under the scalar-path approximation.
Complex cloud paths	Stephens and Heidinger (2000) treats 3-D transport; Bennartz et al. (2006) tests flexible PDF inversion for single and multilayer clouds.	The fitted higher-order polynomial is more flexible than a strict gamma PDF, but it is not a full PDF inversion and cannot uniquely recover multimodal or layer-resolved paths.
Operational fit	Classical inverse-Laplace retrievals are ill conditioned and require high spectral quality (Min et al., 2004; Bennartz et al., 2006).	The project uses a stable low-dimensional summary: exact linear fitting of log transmittance, band-specific truncation, and nonnegative k_1, k_2.
Scientific role	A-band studies use path statistics to diagnose clouds and scattering.	The spectral block is primarily the mechanistic and plume-safety evidence. Feature ablation shows it is approximately neutral for TCCON predictive skill, so physical interpretation should not be overstated as the operational source of correction skill.

11 Manuscript-ready conservative-ensemble derivation

The following compact formulation is recommended for the manuscript. It defines the fitted object unambiguously as the path distribution of the *conservative* reference problem, rather than the absorption-selected distribution at each wavelength.

> Following the equivalence theorem (Irvine, 1964; Partain et al., 2000; Stephens and Heidinger, 2000), we define $P_0(\Gamma)$ as the ensemble of photon trajectories reaching the detector in a conservative reference atmosphere with the same illumination, viewing geometry, surface boundary, and continuum scattering properties as the absorbing atmosphere. Across a sufficiently narrow spectral band, these scattering properties are assumed wavelength independent; wavelength therefore enters only through molecular

absorption. A trajectory Γ contributes at wavelength λ with Beer–Lambert weight

$$w_\lambda(\Gamma) = \exp[-A_\lambda(\Gamma)], \qquad A_\lambda(\Gamma) = \int_\Gamma \alpha_\lambda(\boldsymbol{r})\,\mathrm{d}s. \tag{22}$$

The normalized radiance is consequently

$$\mathcal{T}_\lambda = \mathbb{E}_{P_0}[w_\lambda] = \int P_0(\Gamma)e^{-A_\lambda(\Gamma)}\,\mathrm{d}\Gamma. \tag{23}$$

We approximate the absorber amount along each trajectory by $A_\lambda(\Gamma) \simeq \tau_\lambda \ell'(\Gamma)$, where τ_λ is the modeled direct-slant optical depth and ℓ' is the corresponding normalized path enhancement. If $p_0(\ell')$ denotes the distribution of ℓ' induced by the conservative ensemble P_0, then

$$\mathcal{T}(\tau_\lambda) = \int_0^\infty p_0(\ell')e^{-\tau_\lambda \ell'}\,\mathrm{d}\ell', \tag{24}$$

so the absorption spectrum samples the Laplace transform of one wavelength-independent conservative path distribution. Including an effective continuum reflectance γ gives

$$T_\lambda = \frac{\pi I_\lambda}{F_{0,\lambda}\mu_0} = \gamma\,\mathcal{T}(\tau_\lambda). \tag{25}$$

Expanding the log Laplace transform about zero absorption yields

$$\ln T(\tau) = \ln\gamma + \sum_{n=1}^{\infty}\frac{(-1)^n C_n}{n!}\tau^n \simeq b + \sum_{n=1}^{N}\frac{(-1)^n k_n}{n}\tau^n, \qquad k_n = \frac{C_n}{(n-1)!}, \tag{26}$$

where C_n is the nth cumulant of the conservative distribution $p_0(\ell')$. Hence $k_1 = \mathbb{E}_{p_0}[\ell']$ is the mean path enhancement and $k_2 = \mathrm{Var}_{p_0}(\ell')$ is its variance under the stated approximations.

Immediately after this derivation, the manuscript should include the following qualification:

> The conservative distribution $p_0(\ell')$ is distinct from the path distribution among photons detected within an absorption channel. The latter is $p_{\mathrm{det}}(\ell' \mid \tau) \propto p_0(\ell')e^{-\tau\ell'}$ and therefore varies between weak and strong channels even when the equivalence-theorem assumptions hold. The fitted coefficients describe the common conservative ensemble, inferred from its wavelength- dependent absorption weighting, rather than a separate photon-path distribution at each wavelength.

This language makes the scientific claim testable: the project assumes a common conservative ensemble within each band, not identical detected paths at all optical depths. If layer-dependent absorption cannot be represented by $A_\lambda \simeq \tau_\lambda \ell'$, the coefficients remain band-effective summaries but cease to be exact geometrical-path cumulants.

12 Recommended validation tests

The following tests directly target the assumptions rather than only fit quality:

1. Fit weak and intermediate optical-depth channels, then predict the strong channels. Successful prediction supports one underlying p_0; structured failure identifies either truncation error or a scalar-path failure.

2. Compare channels with similar total τ but different pressure or vertical weighting. A persistent radiance difference after continuum control is evidence that scalar τ is insufficient.

3. Use backward Monte Carlo output to compute both the true geometrical path histogram and layer-resolved absorber amounts. Apply the production estimator to the simulated spectrum and test whether fitted k_1, k_2 close against the corresponding moments.

4. Repeat the closure for independent-column and full 3-D transport. This separates adjacency-driven horizontal paths from ordinary vertical cloud truncation.

5. Examine residuals against wavelength, line identity, pressure broadening, surface type, and polarization after conditioning on τ. Dependence on wavelength beyond τ challenges the common- path assumption.

13 Assumption ledger

Table 2: Assumptions and consequences if they fail.

Assumption	Role	Consequence of failure
Constant scattering properties within a band	Keeps $P_0(\Gamma)$ common to all channels.	The spectrum is not a transform of one underlying trajectory ensemble.
Absorption is passive survival weighting	Establishes the equivalence theorem, equation (2).	Inelastic or coupled processes require a full wavelength-changing transport treatment.
Scalar absorber path, equation (3)	Reduces the trajectory functional to one-dimensional $p_0(\ell')$.	k_1, k_2 become effective absorber- and channel-weighted coefficients rather than geometrical-path moments.
Accurate continuum/reference normalization	Identifies the radiance ratio with transmittance.	Surface or instrument spectral structure leaks into the intercept and cumulants.
Lambertian effective surface	Gives $b = \ln \gamma$.	The intercept is only an effective reflectance; BRDF coupling may also affect curvature.
Adequate Taylor truncation	Connects finite-τ observations to derivatives at $\tau = 0$.	Strong-line information can bias or destabilize the fitted cumulants despite a valid equivalence theorem.
Existence of a physical PDF	Gives cumulant meaning to all coefficients.	A polynomial with $k_1, k_2 \geq 0$ may still not be the log-Laplace transform of any nonnegative PDF.
Gamma PDF, only when invoked	Gives equations (19) and (20).	k_1^2/k_2 is merely gamma-equivalent; independently fitted higher terms need not follow the gamma sequence.

14 Conclusion

The equivalence theorem provides the rigorous starting point: absorption exponentially reweights trajectories generated by an otherwise unchanged scattering problem. The familiar one-dimensional PPDF Laplace transform requires an additional assumption that every trajectory can be summarized by one wavelength-independent absorber-path multiplier. Under that assumption, the project fit has a direct probabilistic interpretation: k_1 is mean path enhancement and k_2 is its variance. Weak and strong channels naturally contain different *detected* path populations, but that alone does not violate the theorem. The important failure mode is wavelength-dependent scattering or layer-dependent absorption that cannot be collapsed onto one scalar path. The current coefficients remain scientifically useful in that case, but should be reported as band-effective path cumulants and validated against layer-resolved Monte Carlo path statistics.

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